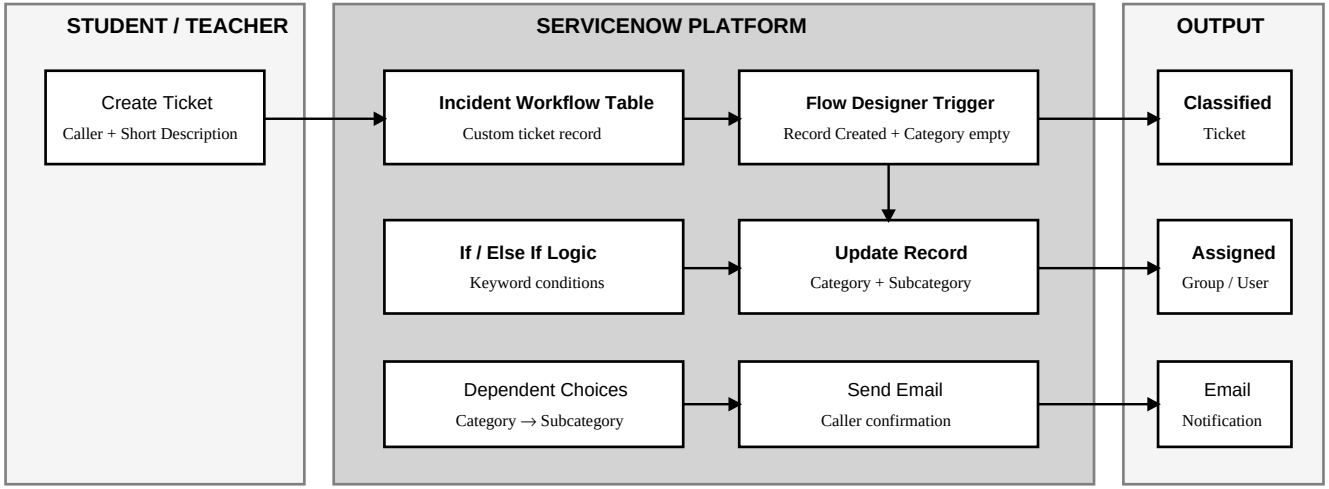
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	03 October 2026
Team ID	SWTID-2026-6328
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	4 Marks

Technical Architecture:

The architecture below is derived from the actual project workflow: a custom **Incident Workflow** table stores ticket records; Flow Designer starts when a record is created with an empty Category, evaluates the Short Description for predefined keywords, updates Category/Subcategory, and sends an email confirmation to the caller. ■filecite■turn1file0■L202-L220■



Actual project flow: ticket creation → keyword-based classification → category/subcategory update → caller email

S.No	Component	Description	Technology
1.	User Interface	Student/teacher or helpdesk user enters Caller and Short Description to create an IT support ticket.	ServiceNow Form / ServiceNow UI

Table-1 : Components & Technologies

S.No	Component	Description	Technology
2.	Application Logic-1	Creates and stores the custom ticket record with auto number, caller, short description, description, state, assignment group and assigned-to fields.	ServiceNow Custom Table — Incident Workflow
3.	Application Logic-2	Starts automation when a new record is created and Category is empty.	Flow Designer — Record Created Trigger
4.	Application Logic-3	Checks Short Description for Wi-Fi/Network, Projector, Password/Login, or Slow/Hanging keywords and updates Category and Subcategory.	Flow Designer — If / Else If conditions + Update Record actions
5.	Database	Stores ticket number, caller, category, subcategory, description, state, assignment group and assigned-to information.	Incident Workflow table / ServiceNow platform data store
6.	Cloud Database	The ticket table and its records are maintained within the ServiceNow platform instance.	ServiceNow Cloud Platform
7.	File Storage	No separate file-storage component is specified in the project. The core workflow stores structured ticket data.	Not required for core implementation
8.	External API-1	No external API is used for the keyword-based classification workflow described in the project.	Not applicable
9.	External API-2	No second external API is specified. Future integrations can be added if required.	Not applicable / future integration
10.	Machine Learning Model	The actual implementation classifies tickets using predefined keyword conditions in Flow Designer; no ML model is specified.	Not applicable — rule-based/no-code classification
11.	Infrastructure (Server / Cloud)	The project is configured and executed inside the ServiceNow instance rather than on a separately developed local application server.	ServiceNow Cloud Platform

Local Server Configuration:

Not required for the implemented ServiceNow workflow.

Cloud Server Configuration:

ServiceNow instance containing the custom Incident Workflow table, configured fields/choices, dependent Subcategory field, Flow Designer flow, and email action.

Project-specific data model:

Field	Type	Reference / Choices
Number	Auto Number	Generated ticket number
Caller	Reference	sys_user
Category	Choice	Network, Hardware, Access, Performance
Subcategory	Choice	Wi-Fi, Projector, Forgot Password, Slow Computer
Short Description	String	Issue summary / keyword source
Description	String	Issue details
State	Choice	New, In progress, On hold, Resolved, Closed
Assigned Group	Reference	sys_user_group
Assigned to	Reference	sys_user

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No separate open-source framework is specified. The solution is implemented as a no-code ServiceNow configuration.	ServiceNow Flow Designer
2.	Security	Access should be limited to authorized users through ServiceNow roles/permissions and table/field access controls. The project also validates that only authorized users can access or modify ticket data.	ServiceNow roles, ACLs, access controls, HTTPS/TLS
3.	Scalable Architecture	The design separates ticket data from automation logic. Additional categories and keyword branches can be added to the Flow Designer flow, while dependent choices keep Category/Subcategory values standardized.	Custom table, Flow Designer, dependent choice configuration
4.	Availability	The workflow executes within the ServiceNow platform. No separate application server is required by the project design.	ServiceNow Cloud Platform
5.	Performance	Classification occurs automatically at ticket creation using simple keyword conditions, reducing manual review. The trigger is restricted to records where Category is empty, avoiding unnecessary reclassification.	Flow Designer trigger, conditions, Update Record actions

Actual Classification Rules

Keyword / condition	Category	Subcategory
WiFi / Network	Network	Wi-Fi
Projector	Hardware	Projector
Password / Login	Access	Forgot Password
Slow / Hanging	Performance	Slow Computer

Flow Designer Configuration

Flow name: Auto Classify School IT Tickets **Trigger:** Record Created **Table:** Incident Workflow **Condition:** Category is Empty.

Actions: If / Else If keyword checks → Update Record with Category and Subcategory → Send Email to Caller.

References:

• Project source document: Auto Ticket Classification using Flow Designer — requirements, configuration, automation, testing, deployment and conclusion. ■filecite■turn1file0■L3-L11■

• Project source document: Custom Incident Workflow table and field/data-type configuration. ■filecite■turn1file0■L78-L91■

• Project source document: Category/Subcategory dependency configuration. ■filecite■turn1file0■L144-L165■

• Project source document: Flow Designer trigger, classification branches and email action. ■filecite■turn1file0■L222-L250■ ■filecite■turn1file0■L256-L325■

Note: This document intentionally describes the implemented rule-based Flow Designer solution. It does not introduce an ML model or external API that is not present in the provided project document.